

第9章 对 $f(t) = t \varepsilon(t)$

$$\begin{aligned}
 9.1 \int_{0-}^{\infty} f(t) e^{-st} dt &= \int_{0-}^{\infty} t \varepsilon(t) e^{-st} dt \\
 &= \int_{0+}^{\infty} t e^{-st} dt = -\frac{1}{s} \int_{0+}^{\infty} t d e^{-st} \\
 &= -\frac{1}{s} \left(t e^{-st} \Big|_{0+}^{\infty} - \int_{0+}^{\infty} e^{-st} dt \right) \\
 &= \frac{1}{s^2}
 \end{aligned}$$

对 $f(t) = t e^{-at} \varepsilon(t)$

$$\begin{aligned}
 \int_{0-}^{\infty} f(t) e^{-st} dt &= \int_{0-}^{\infty} t e^{-(a+s)t} \varepsilon(t) dt \\
 &= \int_{0+}^{\infty} t e^{-(a+s)t} dt = \frac{1}{(a+s)^2}
 \end{aligned}$$

(a)

$$9.2 F(s) = \frac{A}{s+2} + \frac{B}{s+3}$$

$$A = \lim_{s \rightarrow -2} \frac{2s+1}{s^2+5s+6} \cdot (s+2) = -3$$

$$B = \lim_{s \rightarrow -3} \frac{2s+1}{s^2+5s+6} \cdot (s+3) = 5$$

$$\therefore F(s) = \frac{-3}{s+2} + \frac{5}{s+3}$$

$$\therefore \text{原函数 } f(t) = -3e^{-2t} + 5e^{-3t}$$

(b) $F(s) = \frac{s^3+5s^2+9s+7}{s^2+3s+2}$

$$\begin{aligned}
 &= \frac{s+5}{s+2} + \frac{2s^2+7s+7}{s^2+3s+2} \\
 &= s+2 + \frac{s+3}{s^2+3s+2} = s+2 + \frac{A}{s+1} + \frac{B}{s+2}
 \end{aligned}$$

$$A = \lim_{s \rightarrow -1} \frac{s+3}{s+2} = 2$$

$$B = \lim_{s \rightarrow -2} \frac{s+3}{s+1} = -1$$

$$\therefore F(s) = s+2 + \frac{2}{s+1} - \frac{1}{s+2}$$

$$= s+2 + \frac{s+3}{s^2+3s+2} = s+2 + \frac{A}{s+1} + \frac{B}{s+2} \quad \text{原函数为}$$

$$\delta'(t) + 2\delta(t) + 2e^{-t} - e^{-2t}$$

$$(c) F(s) = \frac{3}{[s - (-1 + \sqrt{5}i)][s - (-1 - \sqrt{5}i)]}$$

$$= \frac{A}{s - (-1 + \sqrt{5}i)} + \frac{B}{s - (-1 - \sqrt{5}i)}$$

$$A = \lim_{s \rightarrow (-1 + \sqrt{5}i)} \frac{3}{s - (-1 - \sqrt{5}i)} = \frac{3}{2\sqrt{5}i} = -\frac{3\sqrt{5}}{10}i$$

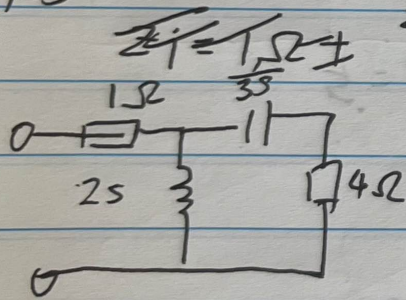
$$B = \lim_{s \rightarrow (-1 - \sqrt{5}i)} \frac{3}{s - (-1 + \sqrt{5}i)} = \frac{3}{-2\sqrt{5}i} = \frac{3\sqrt{5}}{10}i$$

$$f(t) = \therefore \textcircled{1} = -\frac{3\sqrt{5}}{10}i e^{(-1 + \sqrt{5}i)t} + \frac{3\sqrt{5}}{10}i e^{(-1 - \sqrt{5}i)t}$$

$$= -\frac{3\sqrt{5}}{10}i e^{-t} (\cos \sqrt{5}t + i \sin \sqrt{5}t) + \frac{3\sqrt{5}}{10}i e^{-t} (\cos(-\sqrt{5}t) + i \sin(-\sqrt{5}t))$$

$$= \frac{3\sqrt{5}}{5} e^{-t} \sin \sqrt{5}t$$

9.3



$$Z(s) = 1 + \frac{(\frac{1}{3s} + 4\Omega) 2s}{2s + \frac{1}{3s} + 4}$$

$$= 1 + \frac{\frac{2}{3}s + 8s^2}{2s^2 + \frac{1}{3} + 4s}$$

$$= \frac{10s^2 + \frac{14}{3}s + \frac{1}{3}}{2s^2 + \frac{1}{3} + 4s} = \frac{30s^2 + 14s + 1}{6s^2 + 2s + 1}$$

$\therefore$ 为零状态响应

$$9.8 \text{ is } (s) = \frac{1}{s+3}$$

等效阻抗为

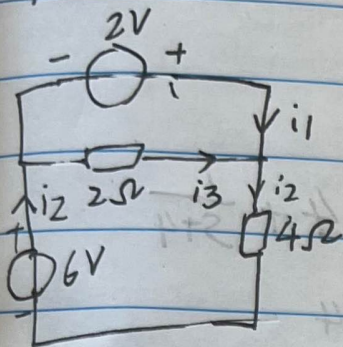
$$Z(s) = \frac{1}{G + \frac{1}{sL} + sC} = \frac{1}{2 + \frac{1}{s} + s} = \frac{s}{s^2 + 2s + 1}$$

$$\therefore U = \textcircled{1} \text{ is } (s) \cdot Z(s) = \frac{s}{(s+3)(s^2 + 2s + 1)} = \frac{\frac{3}{4}}{s+3} + \frac{-\frac{1}{2}}{(s+1)^2} + \frac{\frac{3}{4}}{s+1}$$

$$u(t) = \left(-\frac{3}{4} e^{-3t} - \frac{1}{2} t e^{-t} + \frac{3}{4} e^{-t} \right) \varepsilon(t)$$

$t=0^-$

9.11 稳态时



由 KVL

$$2V + 2\Omega \cdot i_3 = 0$$

$$6V - 2\Omega \cdot i_3 - 4\Omega \cdot i_2 = 0$$

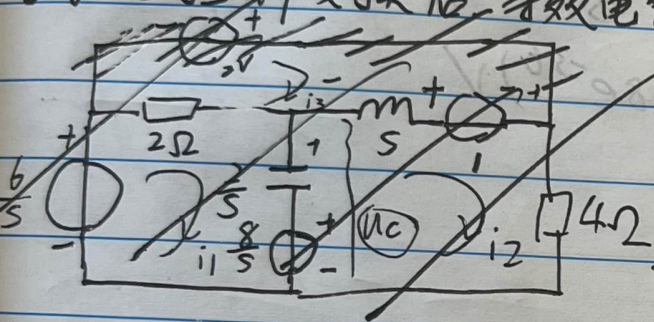
由 KCL, $i_1 + i_3 = i_2$

$$\Rightarrow \begin{cases} i_1 = 3A \\ i_2 = 2A \\ i_3 = -1A \end{cases}$$

即 $i_L(0^-) = i_3 = 1A$

$U_C(0^-) = i_2 \cdot 4\Omega = 8V$

经拉普拉斯变换后等效电路为



由回路电流法

$$(2 + \frac{2}{s})i_1$$

$$\begin{pmatrix} 2 + \frac{2}{s} & \frac{2}{s} & 2 \\ \frac{2}{s} & s+4 & s \\ 2 & s & 2+s \end{pmatrix} \begin{pmatrix} i_1 \\ i_2 \\ i_3 \end{pmatrix} = \begin{pmatrix} -\frac{2}{s} \\ \frac{8}{s}-1 \\ 3 \end{pmatrix}$$

$\Rightarrow i_1 =$

由回路电流法

$$\begin{pmatrix} 2 + \frac{2}{s} & -\frac{2}{s} \\ -\frac{2}{s} & s+4 \end{pmatrix} \begin{pmatrix} i_1 \\ i_2 \end{pmatrix} = \begin{pmatrix} -\frac{2}{s} \\ \frac{8}{s}-1 \end{pmatrix}$$

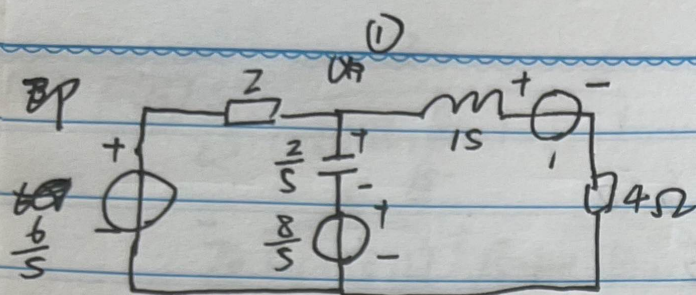
$$\Rightarrow i_1 = \frac{s^2 - 4s}{s^3 + 5s^2 + 5s - 10}$$

$$i_2 = \frac{6 - s^2 + 7s}{s^3 + 5s^2 + 4s - 2}$$

$$i_1 = \frac{\frac{16}{s^2} - \frac{10}{s} - 2}{(2 + \frac{2}{s})(s+4) - \frac{4}{s^2}}$$

$$i_2 = \frac{\frac{12}{s^2} + \frac{14}{s} - 2}{(2 + \frac{2}{s})(s+4) - \frac{4}{s^2}}$$

$$U_C = \frac{2}{s}(i_1 - i_2) + \frac{8}{s}$$



由节点电压法

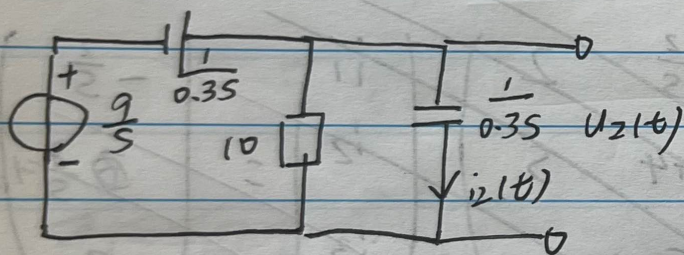
$$u_1 = \left(\frac{1}{2} + \frac{1}{\frac{2}{s}} + \frac{1}{s+4} \right) u_1 = \frac{3}{s} + 4 + \frac{1}{s+4}$$

$$\Rightarrow u_1(s) = \frac{8s^2 + 40s + 24}{s(s+2)(s+3)}$$

$$= \frac{4}{s} + \frac{12}{s+2} - \frac{8}{s+3}$$

$$\therefore u_1 = (4 + 12e^{-2t} - 8e^{-3t})V$$

9.12 拉普拉斯变换后电路图为(单位省略)



$$U_{S2}(s) = \frac{\frac{9}{s}}{\frac{1}{10+0.35s} + \frac{1}{0.35s}}$$

$$= \frac{0.35}{10+0.6s} \cdot \frac{9}{s} = \frac{2.7}{10+0.6s} = \frac{4.5}{s + \frac{100}{6}}$$

$$i_2(s) = \frac{u_2(s)}{0.35} = \frac{0.8/s}{10+0.6s} = 1.35 - \frac{13.5}{0.6s+10} = 1.35 - \frac{22.5}{s + \frac{100}{6}}$$

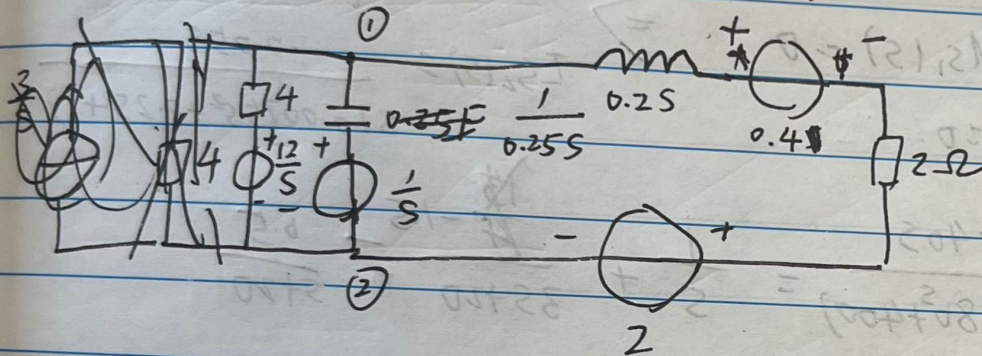
$$\therefore u_2(t) = 4.5e^{-\frac{100}{6}t} \varepsilon(t) \quad i_2(t) = 1.35\varepsilon(t) - 22.5e^{-\frac{100}{6}t} \varepsilon(t)$$

$$U_{S2}(s) = \frac{\frac{1}{\frac{1}{10} + 0.35}}{\frac{1}{\frac{1}{10} + 0.35} + \frac{1}{0.35}} \cdot \frac{9}{s} = \frac{27}{1+6s} = \frac{4.5}{s + \frac{1}{6}}$$

$$I_{S2}(s) = \frac{U_{S2}(s)}{\frac{1}{0.35}} = 1.35 \delta(t) - \frac{0.225}{s + \frac{1}{6}}$$

$$\therefore U_2(t) = 4.5e^{-\frac{1}{6}t} \varepsilon(t) \quad I_2(t) = 1.35\delta(t) - 0.225e^{-\frac{1}{6}t} \varepsilon(t) \text{ (A)}$$

9.14 拉普拉斯变换后电路为(单位省略)



$$U_C = \text{有} \left(\frac{1}{4} + 0.25s + \frac{1}{0.25s+2} \right) U_C = \frac{3}{s} + \frac{\frac{1}{s}}{0.25s} + \frac{2 \times 0.4}{0.25s+2}$$

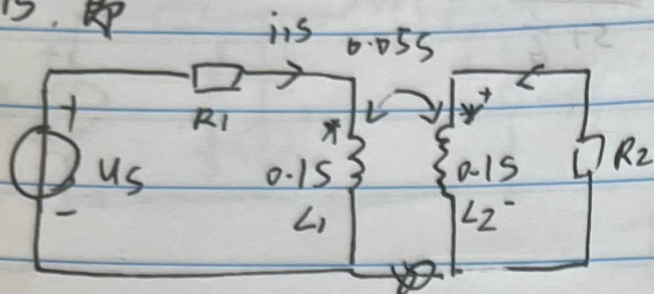
$$\Rightarrow U_C(s) = \frac{9s^2 + 62s + 120}{s(s+5)(s+6)} = \frac{4}{s} + \frac{-7}{s+5} + \frac{12}{s+6}$$

$$\therefore U_C(t) = 4\varepsilon(t) - 7e^{-5t}\varepsilon(t) + 12e^{-6t}\varepsilon(t) \quad \checkmark$$

$$U_C(s) = \frac{s^2 + 54s + 120}{s(s+6)(s+6)} = \frac{4}{s} + \frac{25}{s+5} - \frac{28}{s+6}$$

$$\Rightarrow U_C(t) = 4\varepsilon(t) + 25e^{-5t}\varepsilon(t) - 28e^{-6t}\varepsilon(t) \quad \checkmark$$

拉普拉斯变换后有
9.15. 即



$$U_S(s) = \frac{1}{s}$$

$$U_{S1}(s) = 0.1S I_{S1}(s) + 0.05S I_{S2}(s)$$

$$U_{S2}(s) = 0.1S I_{S1}(s) + 0.1S I_{S2}(s)$$

$$U_S(s) - I_{S1}(s) \cdot R_1 - U_{S1}(s) = 0$$

$$U_{S2}(s) + I_{S2}(s) \cdot R_2 = 0$$

$$\begin{aligned} I_{S1}(s) &= \frac{1 + 0.1S}{s(0.0075s^2 + 0.25s + 1)} \\ I_{S2}(s) &= \frac{-0.05S}{s(0.0075s^2 + 0.25s + 1)} \end{aligned}$$

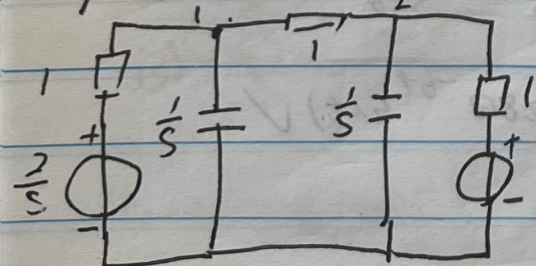
$$I_{S1}(s) = \frac{400 + 40S}{s(35s^2 + 80s + 400)} = \frac{1}{s} + \frac{-1.5}{3s + 20} + \frac{6.5}{s + 20}$$

$$I_{S2}(s) = \frac{-20S}{s(35s^2 + 80s + 400)} = \frac{6.5}{s + 20} - \frac{1.5}{3s + 20}$$

$$I_{S1}(t) = (1 - 0.5e^{-\frac{20}{3}t} + 0.5e^{-20t}) A$$

$$I_{S2}(t) = (0.5e^{-20t} - 0.5e^{-\frac{20}{3}t}) A$$

9.17 拉普拉斯变换后, 即有 由结点电压法



$$(1 + 1 + s)U_1 - 1U_2 = \frac{2}{s}$$

$$-U_1 + (1 + 1 + s)U_2 = 1$$

$$U_1(s) = \frac{3s + 4}{s(s + 1)(s + 3)}$$

$$U_2(s) = \frac{-0.2 + 2 + s}{s(s + 1)(s + 3)}$$

$$U_1(s) = \frac{4}{3} \frac{1}{s} + \frac{-\frac{1}{2}}{s+1} + \frac{-\frac{1}{6}}{s+3}$$

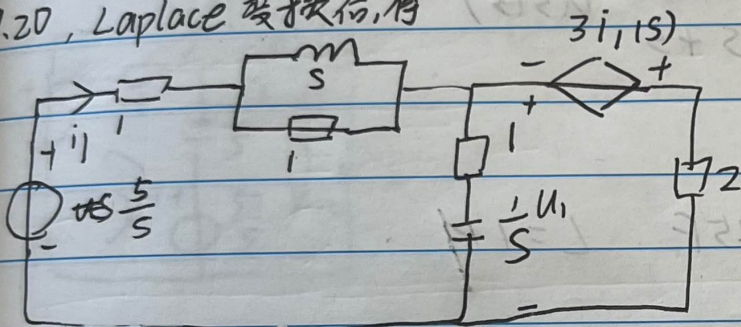
$$U_2(s) = \frac{-\frac{2}{3}}{s} + \frac{-\frac{1}{2}}{s+1} + \frac{\frac{1}{6}}{s+3}$$

$$\Rightarrow -\frac{5}{6}$$

$$\therefore u_1(t) = \frac{4}{3} \varepsilon(t) - \frac{1}{2} e^{-t} \varepsilon(t) - \frac{1}{6} e^{-3t} \varepsilon(t)$$

$$u_2(t) = -\frac{2}{3} \varepsilon(t) - \frac{1}{2} e^{-t} \varepsilon(t) + \frac{1}{6} e^{-3t} \varepsilon(t)$$

9.20, Laplace 变换后, 有



由节点电压法

$$\left(1 + \frac{s}{1+s} + \frac{1}{1+\frac{1}{s}} + \frac{1}{2} \right) u_1(s) = \frac{\frac{5}{s}}{1+\frac{s}{1+s}} - \frac{3i_1(s)}{2}$$

$$\frac{5}{s} + u_1(s) = \left(1 + \frac{s}{1+s} \right) i_1(s) + \frac{5}{s} - \frac{5}{s} = i_1(s) \left(1 + \frac{s}{1+s} \right) + u_1(s)$$

$$\Rightarrow u_1(s) = -\frac{5(1+2s+s^2)}{s^2(3+5s)} = -\frac{1+2s+s^2}{s^2(1+0.6s)}$$

$$= -\frac{0.56}{s} - \frac{1.67}{s^2} - \frac{0.44}{s+0.6}$$

$$\therefore u_1(t) = -0.56 \varepsilon(t) - 1.67 t \varepsilon(t) - 0.44 e^{-0.6t} \varepsilon(t)$$

$$9.23 \text{ } sL \text{ 与 } R_2 \text{ 上电压 } U_1(s) = \frac{\frac{1}{sL + \frac{1}{sL + R_2}}}{R_1 + \frac{1}{sL + \frac{1}{sL + R_2}}} U_s(s)$$

$$= \frac{1}{4Cs + \frac{4}{1+sL} + 1} U_s(s)$$

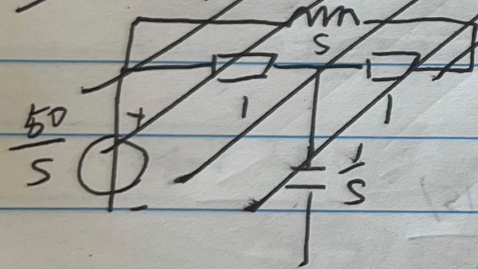
$$I_1(s) = \frac{U_1(s)}{1+sL} = \frac{1}{4Cs(1+sL) + 4 + 1+sL} U_s(s)$$

$$\therefore U(s) = \frac{1}{4CLs^2 + (4C+L)s + 5} U_s(s)$$

$$\text{即 } 4CL = 1 \quad 4C + L = 2$$

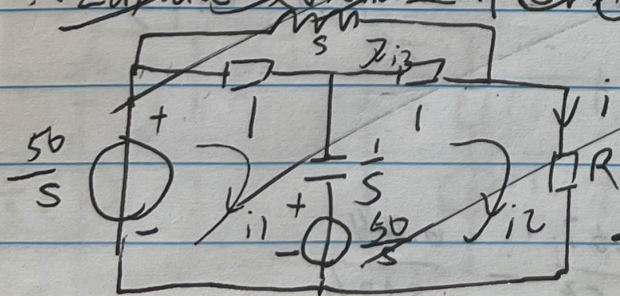
$$\because C > 0 \quad L > 0 \Rightarrow C = 0.25F \quad L = 1H$$

9.26 Laplace 变换后, 得电路图



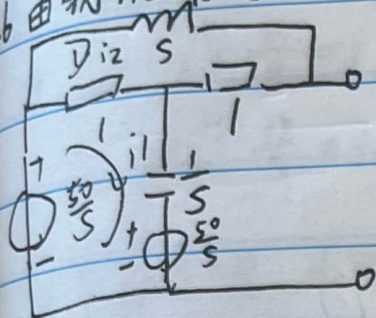
9.26 $t=0^-$ 时, $i_L = 0$ $u_C = 50V$

$\therefore$ Laplace 变换后运算电路图为



$$\begin{bmatrix} (1+\frac{1}{s})i_1 \\ 1+\frac{1}{s} & -\frac{1}{s} & -1 \\ -\frac{1}{s} & 1+R+\frac{1}{s} & -1 \\ -1 & -1 & 2+s \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 50 \\ 0 \end{bmatrix}$$

6 由戴维南定理与 Laplace 变换, 开路时



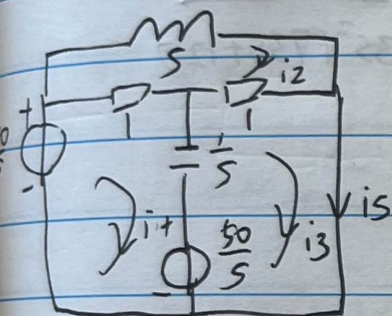
由回路电流法

$$\text{有 } (1 + \frac{1}{s})i_1 - i_2 = 0$$

$$(1 - i_1) + (2 + s)i_2 = 0$$

$$\Rightarrow i_1 = 0 \quad i_2 = 0 \quad \therefore \text{开路电压 } U = \frac{50}{s}$$

短路时, 有



由回路电流法

$$(1 + \frac{1}{s})i_1 - i_2 - \frac{1}{s}i_3 = 0$$

$$-i_1 + (2 + s)i_2 - i_3 = 0$$

$$-\frac{1}{s}i_1 - i_2 + (1 + \frac{1}{s})i_3 = \frac{50}{s}$$

$$\Rightarrow i_3 = i = \frac{50s^2 + 100s + 100}{s^2(2s + 2)}$$

$$\therefore Z = \frac{U}{i} = \frac{50}{s^2 + 2s + 2}$$

$$i = \frac{U}{Z + R} = \frac{\frac{50}{s}}{\frac{50}{s^2 + 2s + 2} + 1} = \frac{50s^2 + 100s + 100}{s^2(2s + 2)}$$

$$\therefore \text{总阻抗为 } Z + R = \frac{(1 + R)^2 + (2R + 2)s + R^2}{s^2 + 2s + 2}$$

$$= \frac{25s^2 + 50s + 50}{2s(s + 1)^2}$$

$$\text{当 } \Delta = 4R^2 + 8R + 4 - 8R^2 - 8R = 0 \quad \text{即当 } R = 1 \Omega \text{ 时,}$$

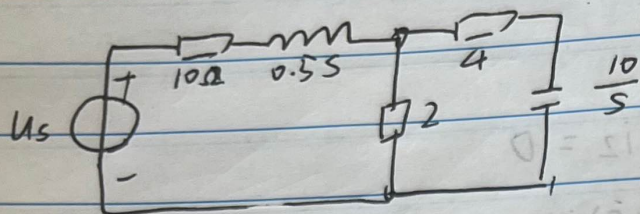
为临界状态

$$= \frac{50}{s} + \frac{-25}{(s + 1)^2} - \frac{25}{s + 1}$$

$$\text{此时 } i = \frac{U}{Z + R} = \frac{\frac{50}{s}}{\frac{50}{s^2 + 2s + 2} + 1} = \frac{50s^2 + 100s + 100}{s^2(2s + 2)} = \frac{25}{s} + \frac{25}{s + 1} - \frac{50}{(s + 1)^2}$$

$$\therefore i = 25 + 25e^{-2t} - 50te^{-2t} \quad (\text{A}) \quad \text{即 } i(t) = 50 - 25te^{-t} - 25e^{-t} \quad (\text{A})$$

9.28 拉普拉斯变换得运算电路



由节点电压法有
$$\left(\frac{1}{10+0.5S} + \frac{1}{4+\frac{10}{S}} + \frac{1}{2} \right) U_1 = \frac{U_S(s)}{10+0.5S}$$

则
$$H(s) = \frac{U_1(s)}{U_S(s)} = \frac{1}{1 + \frac{10+0.5S}{4+\frac{10}{S}} + \frac{1}{2}} = \frac{8S+20}{3S^2+73S+120}$$

当 $U_S = 100\sqrt{2} \cos(10t) \text{ V}$ 时

$$U_S(s) = 100\sqrt{2} \left(\frac{S}{S^2+100} \right)$$

$$U_1(s) = \frac{8S+20}{3S^2+73S+120} \cdot \frac{100\sqrt{2} S}{S^2+100}$$

$$= 100\sqrt{2} \frac{8S^2+20S}{3S^4+73S^3+120S^2+1300S^2+7300S+12000}$$

$$= 100\sqrt{2} \frac{8S^2+20S}{3S^4+73S^3+1420S^2+7300S+12000}$$

$$H(j\omega) = \frac{8j\omega+20}{-3\omega^2+73j\omega+120}$$

$\omega = 10 \text{ rad/s}$ 代入

$$H(j\omega) = \frac{80j+20}{-300+730j+120}$$

$$U_S = 100\sqrt{2} \angle 0^\circ \text{ V}$$

$$\therefore U = U_S \cdot H(j\omega) = 10.97 \angle -27.88^\circ \text{ V}$$

$$\therefore u = 10.97\sqrt{2} \cos(10t-27.88^\circ) \text{ V}$$

9.29 Laplace 变换后. 即

当 $U_S(s) = \frac{6}{s}$ 时

$$U_O(s) = \frac{8}{s} + \frac{2}{s-0.2}$$

$U_S(s) = \frac{12}{s}$ 时

$$U_O(s) = \frac{11}{s} - \frac{1}{s-0.2}$$

由叠加定理, 为零状态响应 + 零输入响应

$$\text{零状态响应有 } H(s) = \frac{U_1(s)}{U_S(s)}$$

$$\therefore U_O(s) = H(s)U_S(s) + U_2(s)$$

$$\Rightarrow \frac{8}{s} + \frac{2}{s-0.2} = H(s) \cdot \frac{6}{s} + U_2(s)$$

$$\frac{11}{s} - \frac{1}{s-0.2} = H(s) \cdot \frac{12}{s} + U_2(s)$$

$$\Rightarrow U_2(s) = \frac{5}{s} + \frac{5}{s-0.2} \quad H(s) = \frac{1}{2} - \frac{3s}{6s-1.2}$$

$$\therefore \text{当 } U_S = 6e^{-5t} \varepsilon(t) \quad \text{即 } U_S(s) = \frac{6}{s-5} \text{ 时}$$

$$U_{10}(s) = \frac{3}{s-5} - \frac{18s}{(6s-1.2)(s-5)} + \frac{5}{s} + \frac{5}{s-0.2}$$

$$= \frac{3}{s-5} - \frac{18s}{(5s-1)(s-5)} + \frac{5}{s} + \frac{5}{s-0.2}$$

$$= \frac{3}{s-5} + \frac{0.125}{s-0.2} - \frac{3.125}{s-5} + \frac{5}{s} + \frac{5}{s-0.2}$$

$$= \frac{-0.125}{s-5} + \frac{5.125}{s-0.2} + \frac{5}{s}$$

$$\therefore U_O = (-0.125e^{-5t} + 5.125e^{-0.2t} + 5) \varepsilon(t) \checkmark$$

$$9.3) H(j\omega) = \frac{U_o(j\omega)}{U_i(j\omega)} \quad \text{代入 } \omega = 4 \text{ rad/s}$$

$$\Rightarrow H(j\omega) = \frac{1}{(4j+1)(4j+2)}$$

$$\therefore \text{稳态分量 } U_o = H(j\omega) \cdot U_i = 2 \angle 0^\circ \text{ V}$$

$$\therefore U_o = 2\sqrt{2} \cos(4t)$$

由 $H(s)$ 形式, 其自由分量形成 $Ae^{-t} + Be^{-2t}$

$$\text{设 } U_o = 2\sqrt{2} \cos(4t) + Ae^{-t} + Be^{-2t}$$

$$\text{代入 } U_o(0+) = 0 \quad \left. \frac{dU_o}{dt} \right|_{t=0+} = 0$$

$$\Rightarrow 2\sqrt{2} + A + B = 0$$

$$2\sqrt{2} - A - 2B = 0$$

$$\Rightarrow B = 2\sqrt{2} \quad A = -4\sqrt{2}$$

$$\Rightarrow U_o = 2\sqrt{2} \cos(4t) - 4\sqrt{2}e^{-t} + 2\sqrt{2}e^{-2t} \text{ (V)}$$